

House Price Prediction using Linear Regression

Artificial Intelligence & Machine Learning – Task 1

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1. Objective

Build and evaluate a Linear Regression model using the California Housing Dataset to predict median house prices.

2. Dataset Description

The California Housing Dataset contains 20,640 samples with 8 input features and one target variable (MedHouseVal). No missing values were found.

3. Exploratory Data Analysis (EDA)

EDA included dataset inspection, summary statistics, missing value analysis, histograms, a correlation heatmap, and a scatter plot between Median Income and Median House Value. (Insert screenshots from your notebook here.)

4. Model Building

The dataset was split into training (80%) and testing (20%) sets. A Linear Regression model was trained using scikit-learn.

5. Model Evaluation

Mean Absolute Error (MAE): **0.5332**

Root Mean Squared Error (RMSE): **0.7456**

R² Score: **0.5758**

6. Conclusion

The Linear Regression model successfully predicted house prices. EDA helped understand feature relationships, and the model achieved satisfactory performance. More advanced algorithms can further improve accuracy.